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# LLM COMPARISON: Gemini | GPT-2 (local) | Mistral (HF Inference API)
# -----
# !pip install -q google-generativeai transformers huggingface_hub

# -----
# 1. IMPORTS
# -----
import google.generativeai as genai
from transformers import pipeline
from huggingface_hub import InferenceClient

# -----
# 2. API KEYS
# -----
GEMINI_API_KEY = ""
HF_TOKEN       = ""

genai.configure(api_key=GEMINI_API_KEY)
hf_client = InferenceClient(token=HF_TOKEN)

# -----
# 3. LOAD LOCAL GPT-2
# -----
print("Loading GPT-2 locally (small, ~500MB)...")
gpt2_pipeline = pipeline("text-generation", model="gpt2")
print("GPT-2 ready.\n")

# -----
# 4. RESPONSE FUNCTIONS
# -----
def get_gemini_response(prompt):
    """Gemini via your API key."""
    model = genai.GenerativeModel("gemini-1.5-flash")
    response = model.generate_content(prompt)
    return response.text.strip()

def get_gpt2_response(prompt):
    """GPT-2 runs fully locally – no API key needed."""
    out = gpt2_pipeline(prompt, max_new_tokens=60,
num_return_sequences=1, do_sample=True)
    # Strip the original prompt from the output
    return out[0]["generated_text"][len(prompt):].strip()

def get_mistral_response(prompt):
    """Mistral 7B via HuggingFace free Inference API – no local
download."""
    response = hf_client.chat_completion(
        model="mistralai/Mistral-7B-Instruct-v0.2",
        messages=[{"role": "user", "content": prompt}],

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        max_tokens=100,
    )
    return response.choices[0].message.content.strip()

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# 5. PROMPT
# -----
prompt = "Write a haiku about AI and nature."

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# 6. GET RESPONSES
# -----
print("Querying models...\n")

results = {}

try:
    results["Gemini 1.5 Flash"] = get_gemini_response(prompt)
    print("✅ Gemini done")
except Exception as e:
    results["Gemini 1.5 Flash"] = f"[Error: {e}]"

try:
    results["GPT-2 (local)"] = get_gpt2_response(prompt)
    print("✅ GPT-2 done")
except Exception as e:
    results["GPT-2 (local)"] = f"[Error: {e}]"

try:
    results["Mistral 7B (HF API)"] = get_mistral_response(prompt)
    print("✅ Mistral done")
except Exception as e:
    results["Mistral 7B (HF API)"] = f"[Error: {e}]"

# -----
# 7. EVALUATION
# -----
def evaluate(text):
    """Score based on uniqueness, low repetition, haiku-like
    brevity."""
    words = text.split()
    if not words:
        return 0
    unique_words = len(set(words))
    repeats = len(words) - unique_words
    length_score = max(0, 15 - abs(len(words) - 12)) # ~12 words ≈
    haiku length
    return unique_words + length_score - (repeats * 2)

scores = {model: evaluate(output) for model, output in

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results.items()}\nranked = sorted(scores.items(), key=lambda x: x[1], reverse=True)
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# 8. PRINT RESULTS
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print("\n" + "="*50)\nprint("  LLM Comparative Analysis")\nprint("="*50)
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for model, output in results.items():\n    print(f"\n {model}")\n    print(f"    Prompt : {prompt}")\n    print(f"    Response:\n        {output}\n")\n    print("-"*50)
```

```
print("\n Ranking (Best → Worst)\n")\nfor i, (model, score) in enumerate(ranked, start=1):\n    print(f" {i}. {model} - Score: {score}")
```

Loading GPT-2 locally (small, ~500MB)...

```
{"model_id": "44b86ec8229849dea4f4bea020046fe6", "version_major": 2, "version_minor": 0}
```

GPT2LMHeadModel LOAD REPORT from: gpt2

| Key                  | Status     |  |
|----------------------|------------|--|
| h.{0...11}.attn.bias | UNEXPECTED |  |

Notes:

- UNEXPECTED : can be ignored when loading from different task/architecture; not ok if you expect identical arch.

GPT-2 ready.

Querying models...

WARNING:tornado.access:404 POST /v1beta/models/gemini-1.5-flash:generateContent?%24alt=json%3Benum-encoding%3Dint (::1) 508.25ms  
Passing `generation\_config` together with generation-related arguments=({'num\_return\_sequences', 'max\_new\_tokens', 'do\_sample'}) is deprecated and will be removed in future versions. Please pass either a `generation\_config` object OR all generation parameters explicitly, but not both.  
Setting `pad\_token\_id` to `eos\_token\_id`:50256 for open-end generation.  
Both `max\_new\_tokens` (=60) and `max\_length` (=50) seem to have been set. `max\_new\_tokens` will take precedence. Please refer to the documentation for more information.

([https://huggingface.co/docs/transformers/main/en/main\\_classes/text\\_generation](https://huggingface.co/docs/transformers/main/en/main_classes/text_generation))

□ GPT-2 done

## □ LLM Comparative Analysis

□ Gemini 1.5 Flash

Prompt : Write a haiku about AI and nature.

Response:

[Error: 404 POST

<https://generativelanguage.googleapis.com/v1beta/models/gemini-1.5-flash:generateContent?%24alt=json%3Benum-encoding%3Dint>:  
models/gemini-1.5-flash is not found for API version v1beta, or is not supported for generateContent. Call ListModels to see the list of available models and their supported methods.]

□ GPT-2 (local)

Prompt : Write a haiku about AI and nature.

Response:

It's a simple script with a simple syntax that uses a bit of syntax to express the idea of our world.

The simplest way of doing this is to use a simple syntax such as <script> {...}, where <script> is a set of code. So for example.

□ Mistral 7B (HF API)

Prompt : Write a haiku about AI and nature.

Response:

[Error: (Request ID: Root=1-699d1dca-316719dc545c02bc072cc63a;6b008e70-e50c-448b-96b3-abbefeacd0a5)

403 Forbidden: This authentication method does not have sufficient permissions to call Inference Providers on behalf of user mightyraj. Cannot access content at:  
<https://router.huggingface.co/v1/chat/completions>.  
Make sure your token has the correct permissions.]

□ Ranking (Best → Worst)

1. Mistral 7B (HF API) – Score: 36
2. Gemini 1.5 Flash – Score: 19
3. GPT-2 (local) – Score: 5

